

BHAVYA DHAKA

Dublin, Ireland | bhavya.dhaka@ucdconnect.ie | linkedin.com/in/bhavyadhaka | github.com/bhavya1929 | Work Authorization: Ireland (STAMP 4)

PROFESSIONAL SUMMARY

Computational biologist with a strong background in machine learning, cancer genomics, and perturbation biology. Seven years of experience applying deep learning and single-cell multi-omics to understand how genes, drugs and tumours behave and how those insights translate into therapeutic decisions. Designed and built novel model architectures end-to-end, leading multiple projects independently from conception through to publication. Contributed to large-scale international consortia and collaborated with multidisciplinary research groups across institutions in India, Ireland, and the United States. Published in high-impact peer-reviewed journals with extensive experience across academic and international research environments, combining computational and biological expertise and modern AI approaches to drive drug discovery and translational insight.

CORE SKILLS

AI / ML: PyTorch, deep learning, meta-learning, graph neural networks, transformer fine-tuning, perturbation modelling, representation learning, sequence modelling, foundation model benchmarking, model architecture design and optimisation, working knowledge of agentic AI frameworks (LangChain, AutoGen)

Genomics & Single-Cell: scRNA-seq, Perturb-seq, CRISPR screens, in silico screening, ATAC-seq, ChIP-seq, RNA-seq, multi-omics integration, CNV analysis, functional genomics, Scanpy, Seurat

Tools & Infrastructure: Python, R, Bash, SQL, Nextflow, Docker, SLURM/HPC, Git, CloudCIX, reproducible pipeline development

Domain Knowledge: Perturbation biology, cancer genomics, drug response prediction, therapeutic target prioritisation, precision oncology, regulatory genomics, biomarker discovery

EXPERIENCE

Machine Learning Researcher | Ideker Lab, UC San Diego, USA

2024 – 2026

- **Developed HyperMap**, a scalable PyTorch meta-learning framework for predicting gene and drug perturbation responses, achieving ~70% improvement over leading transformer-based models with strong cross-context generalisation across unseen cell lines and disease settings.
- **Running ~20x faster** than prior approaches, enabling high-throughput in silico prioritisation of perturbations and reducing experimental burden for downstream target discovery workflows.
- **Generated 370+ novel genetic perturbation profiles**, expanding perturbation atlases to support scalable modelling of therapeutic hypotheses across cancer and rare disease contexts.
- **Designed the system for accessibility and reuse**, usability for labs with limited computational infrastructure and broadening translational reach of the framework.

Computational Scientist (MSCA Doctoral Fellow) | GoldLab, University College Dublin

2022 – Present

Marie Skłodowska-Curie Actions Fellowship - competitive EU-wide award

- **Extended TransCistor**, a cis-regulatory lncRNA discovery framework, to characterise 268 human and 134 mouse regulatory targets, enabling systematic inclusion of non-coding elements in CRISPR study designs and perturbation atlases.
- **Contributed multi-omics analytical expertise to the FANTOM6 consortium** (50+ institutions globally), profiling RNA–DNA interactions across cell types in a large-scale regulatory genomics study currently under review at Nature.
- **Mentored MSc and PhD students** across computational genomics and machine learning projects, guiding multiple students to project completion, providing hands-on supervision experience across academic levels.
- **Organised a 150-person national genomics symposium** and secured industry sponsorship; chaired a monthly international speaker series; to create community engagement.

- **Led two independent cancer genomics projects:** chromatin accessibility profiling in p53 gain-of-function mutants and pan-cancer immunoproteasome activity analysis, identifying regulatory mechanisms directly relevant to immunotherapy response and tumour stratification.
- **Built reproducible Nextflow pipelines processing 500+ tumour samples** for ATAC-seq and RNA-seq workflows, achieving ~40% runtime reduction and enabling scalable, version-controlled pan-cancer regulatory analysis.
- **Collaborated with NIH** on scRNA-seq analysis of small cell lung cancer patient cohorts, modelling tumour evolution and transcriptional heterogeneity to support translational clinical research.

PUBLICATIONS

1. **Dhaka B** et al. HyperMap: An Efficient Meta-Learning Framework for Transferring Perturbation Responses Across Diverse Biological Contexts. (Preprint, 2026)
2. Lambolez A, ..., **Dhaka B**, ..., Carninci P et al. The human RNA-DNA interactome is cell type-specific and dynamic. Nature (Under review, 2026)
3. **Dhaka B** et al. Functional identification of cis-regulatory long noncoding RNAs at controlled false-discovery rates. Nucleic Acids Research (2024)
4. Kumar R*, **Dhaka B*** et al. Prognostic association of immunoproteasome expression in solid tumours. Molecular Oncology (2023) [Co-first author]
5. **Dhaka B** & Sabarinathan R. Differential chromatin accessibility landscape of gain-of-function mutant p53 tumours. BMC Cancer (2021)

SELECTED TALKS

- **2026 Perturb2026 Conference** — Selected Talk
- **2025 EMBL Conference: Cancer Genomics** — Selected Talk

AWARDS & FELLOWSHIPS

- **2022 Marie Skłodowska-Curie Actions (MSCA) PhD Fellowship** — Competitive EU-wide doctoral fellowship awarded to outstanding early-stage researchers
- 2022 Indian Council of Medical Research (ICMR) Senior Research Fellowship
- 2019 Bangalore Life Science Cluster (BLiSC) Fellowship

EDUCATION

PhD, Genomics Data Science — University College Dublin, Ireland

Expected August 2026

Thesis: "From Predictive Models to Mechanistic Insight in Perturbation Biology"

MSc, Bioinformatics & Applied Biotechnology — IBAB, India